IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

SMART GRID AND DATA SCIENCE

Friday, 8 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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The Questions

Question 1 (20 marks)

In an electricity market, three conventional generation units and one renewable wind farm participate in market clearing. The operational characteristics of the generation units are given in Table 1. Three demand participants submit price–quantity bids to the market, as shown in Table 2.

Table 1: Operational characteristics of generation units

Unit	R1	U1	U2	U3
Technology	Wind	Coal	Gas	Gas
Cost (£/MWh)	0.0	9.8	14.1	20.7
Capacity (MW)	300	200	400	120

Table 2: Demand bids

Demand	D1	D2	D3
Price bid (£/MWh)	32.5	26.4	18.5
Quantity bid (MW)	200	500	100

a) During a period of strong wind conditions, the wind farm "R1" operates at 80% of its rated capacity. Draw the market demand-supply curves and determine the market-clearing price. Calculate the generation dispatch and profit of each generation unit (including the wind farm) and the demand benefit of each demand participant.

[8]

b) Assume the market experiences a period of low wind conditions, such that wind farm "R1" operates at 20% of its capacity. Draw the updated market demand-supply curve and determine the new market-clearing price. Recalculate the generation dispatch and profit of each generation unit (including the wind farm) and the demand benefit of each demand participant.

[8]

c) An energy storage unit "S1" participates in the market to perform energy arbitrage, i.e., charging during low-price period and discharging during high-price period. The operational characteristics of the storage unit are given in Table 3. Assuming the battery can only change and discargh once, determine the optimal charging and discharging power schedule, and the market revenue obtained from energy arbitrage.

[4]

Table 3: Operational characteristics of the energy storage unit

Unit	S1
Initial energy (MWh)	5
Charging power capacity (MW)	10
Discharging power capacity (MW)	10
Energy capacity (MWh)	20

Question 2 (20 marks)

a) In the context of wind generation forecasting, what does zero-shot forecasting mean? Define auto-regressive forecasting and contrast it with context/covariate-enhanced forecasting.

[5]

b) Give two concrete examples of data leakage that commonly happen in time-series forecasting pipeline for power systems. For one example, state one mitigation.

(Hint: *Any step that uses information from* ***validation/test timestamps*** *during training is leakage).*

[3]

c) Assume that you are a responsible system operator who cares more about large under-forecasts than over-forecasts. Answer the following questions.

i) Name one standard regression loss and why it may be misaligned.

[2]

ii) For a wind farm, the actual power at $t+1$ is $y = 0.40$p.u. A model predicts the $\tau = 0.9$ quantile $\hat{y} = 0.35$p.u. Compute the pinball loss using the equation below. What is the advantages of pinball loss for wind forecasting?

$$\mathcal{L}_\tau(y, \hat{y}) = \begin{cases} \tau(y - \hat{y}), & y \geq \hat{y} \\ (1-\tau)(\hat{y} - y), & y < \hat{y} \end{cases}$$

[2]

iii) Day-ahead schedule uses median forecast $q_{0.5} = 0.50$p.u. and risk margin uses $q_{0.1} = 0.35$p.u. Define reserve $R = q_{0.5} - q_{0.1}$. Compute R and interpret it.

[2]

d) Predict next-hour net load with $\hat{y} = wx$. For one data point: $x = 2$, $y = 10$. Initial $w = 4$. Loss $\ell = \frac{1}{2}(\hat{y} - y)^2$. Learning rate $\eta = 0.1$. Compute one SGD update.

[6]

Question 3 (20 marks)

a) Monte Carlo simulation is applied to estimate the expected number of interruptions.

i) The application of Monte Carlo simulations relies on three fundamental theories. Please state these three theories and explain how they support the application.

[3]

ii) After N=10,000 samples the best estimate of the interruption is 115,000 MW, with a standard error of 2093 MW. How many samples will be required (approximately) to reduce the coefficient of variation to 0.26%?

[3]

b) For a cyber-physical power system,

i) State examples of physical threats and cyber threats to reliable power system operation. For each category, provide one example of an accidental/natural threat and one example of a malicious threat (i.e. provide four examples in total).

[3]

ii) State the general idea or approach to detect a cyber attack.

[2]

iii) A detector is evaluated on 200 samples: 40 attacks and 160 normal. It predicts 30 attacks correctly, misses 10 attacks, and raises 20 false alarms. Compute the True Positive Rate (TPR) and False Positive Rate (FPR) of this detector.

[2]

c) For the distribution network with high penetration of smart meter at the household level,

i) State at least three examples of key information can be identified through smart meter data.

[3]

ii) List at least three Privacy Enhancing Technologies (PETs) and discuss their suitability for smart metering system.

[4]

Question 4 (20 marks)

Consider the following power system with 4 buses and 5 branches. The branch reactance and power flow measurements are given by the tables below. The per unit base is given as 100MVA. Bus 1 is set as the slack bus. Answer the questions below.
Note: By convention, $\boldsymbol{P_{12}}$ represents the power flow from bus 1 to bus 2.

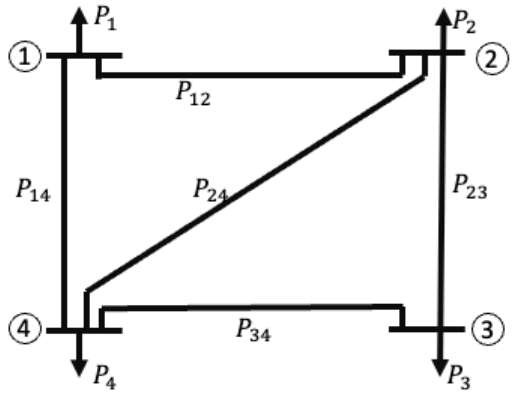


Figure 4-bus system with branch reactance and measurements

Line reactance (in p.u.):		**Power flow measurement (MW)**	
x_{12}	0.4	P_{12}	50
x_{14}	0.1	P_{14}	500
x_{23}	0.2	P_{23}	-50
x_{24}	0.2	P_{24}	150
x_{34}	0.5	P_{34}	80

a) The incidence matrix of above power system is written as follows. According to the incidence matrix, construct the measurement matrix **H** for this system under the DC assumptions. What is the rank of **H** and is the system observable? Why?

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

[4]

b) Find the power injection on each bus. Which bus is most likely to be a load bus?

[3]

c) Given the power flow measurements and the measurement matrix **H** in a), find the current state of the system and the residual of the measurement under the noiseless condition.

[3]

d) Given that the load flow limit on line 1-4 is 800MW, an attacker would like to convince the system operator that line 1-4 is overloaded, e.g. to modify the measurement $\mathbf{P_{14}}$ from 500MW to 800MW. Is it possible to only change $\mathbf{P_{14}}$ without being detected by the bad data detection (BDD) and why? Assume the attacker only changes the voltage angle at bus 4, calculate the attack vector that can bypass the BDD under the noiseless condition.

[5]

e) A residual-based BDD uses $J = r^\top R^{-1} r$ for anomaly detection. Assume Gaussian measurement noise and correct model.

i) State the approximate distribution of J under no attack.

[1]

ii) For this system, what is the degree of freedom?

[1]

f) Briefly describe the concept of Moving Target Defence (MTD) against False Data Injection Attacks (FDIA). Based on the attack formulated in d), determine which transmission line reactance should be modified if only one can be changed.

(Note: there may be multiple answers to this question.)

[3]